LLM-Driven Multimodal Knowledge Graph Construction for Industrial Process With Prompt Optimization and Fuzzy RAG

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***Abstract*—The industrial flotation process involves multimodal data and cross-procedural knowledge, presenting significant challenges for knowledge system management and traditional knowledge graph (KG) construction methods. This study develops a large language model (LLM)-driven framework to construct a domain-specific multimodal KG. The flotation knowledge tree ontology is designed as the structural backbone of the KG to organize multilevel operations and interactions while integrating heterogeneous flotation data. To enhance the LLM's domain adaptability, adaptive prompt optimization is proposed to iteratively refine the extraction template with flotation-specific examples and performance feedback, enabling more accurate triple extraction. In addition, a fuzzy information entropy-driven retrieval-augmented generation (RAG) method is proposed, leveraging fuzzy logic to assign weights to key terms and numerical contexts to preserve causal relationships and semantic integrity under data uncertainty. Furthermore, we established a two-stage LLM-driven pipeline to generate initial triples using a low-rank adaptation (LoRA) fine-tuned lightweight LLM with the optimized prompt and fuzzy RAG context, followed by refinement with a large-scale LLM for precision and ontology alignment. Validated triples are mapped onto the flotation knowledge tree to form a structured, high-quality KG with minimal manual intervention. This framework fuses ontology structuring, fuzzy semantic retrieval, and LLM reasoning to enable automated, domain-tailored knowledge assembly for industrial flotation processes. The constructed KG is evaluated using the flotation knowledge tree-based method, achieving a score of 0.92, with a structural score of 0.97, and a semantic score of 0.88, demonstrating robustness and coherence.**

***Index Terms*—Adaptive prompt optimization, flotation knowledge tree, large language model (LLM), multimodal knowledge graph (KG), retrieval-augmented generation (RAG).**

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I. Introduction

FROTH flotation, a cornerstone of mineral processing, employs air bubbles to separate hydrophobic minerals from hydrophilic ones and is widely used in extracting metals and nonmetals, such as copper, lead, zinc, sulfur, and coal [1], [2], [3]. This process is inherently complex due to its intricate mechanisms and the involvement of multiple interconnected flotation cells. These cells are coupled through froth and pulp flows, forming a complex network topology [4]. Furthermore, its knowledge system spans interconnected processes, from ore pretreatment to concentrate recovery, encompassing multimodal data (e.g., structured inspection records, unstructured images, and text) and cross-procedural knowledge, including parameter correlations across cells and reagent addition decisions [5]. Traditional methods struggle to integrate this multimodal data effectively, posing significant challenges to constructing a knowledge graph (KG) for industrial flotation processes.

KGs provide a robust framework for structured knowledge representation and have gained significant traction in industrial applications, particularly in geology and mineral processing, where they facilitate geological report analysis and automated resource estimation, thereby enhancing operational efficiency [6], [7]. Traditional KG construction methods typically rely on manual curation or rule-based extraction, which are well-suited for structured data sources. However, when addressing the vast, heterogeneous data generated by industrial processes, such as froth flotation, these methods are labor-intensive and lack scalability [8]. Recent advancements have utilized semantic data management techniques to integrate diverse data sources [9] and developed efficient KG embedding methods to tackle these challenges [10]. Nevertheless, these approaches primarily focus on textual data, offering limited support for integrating multimodal data, such as images.

In recent years, the rapid advancement of large language models (LLMs), such as GPT-4 and DeepSeek, has opened new opportunities for KG construction [11]. LLMs excel at extracting entities and relationships from unstructured text, thereby automating the KG construction process [12], [13], [14]. Moreover, by analyzing textual descriptions of images, LLMs enable the integration of image-related information into KGs, providing a novel approach to multimodal knowledge management in froth flotation. In engineering, LLMs have been utilized to optimize designs and model complex systems [15], [16], demonstrating

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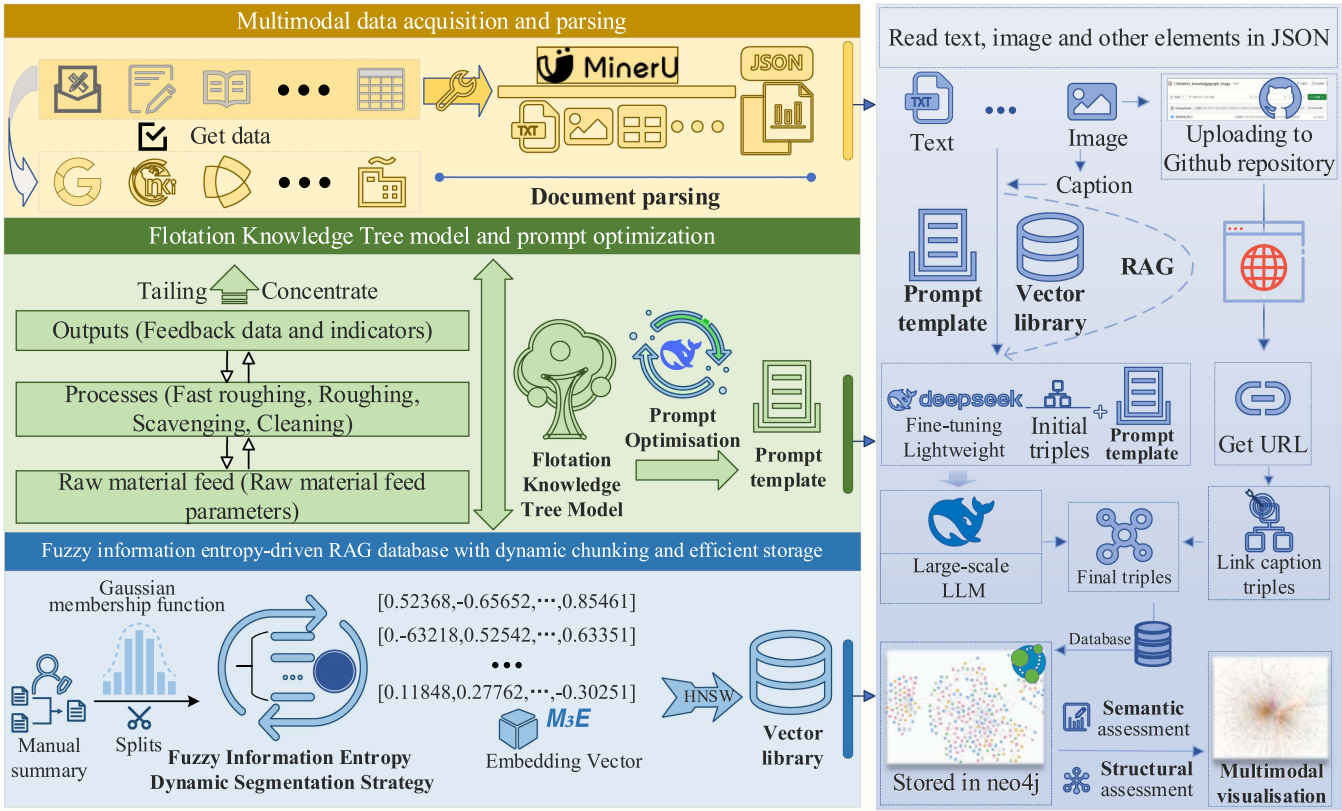


Fig. 1. LLM-driven multimodal KG construction framework for flotation processes.

significant potential for industrial applications [17], [18]. Pan et al. proposed a unified framework for integrating LLMs and KGs, highlighting their synergistic role in industrial knowledge management [12], [13]. Zou et al. [14] developed automated text-to-KG conversion using LLMs. Tan et al. explored LLM applications in KG reasoning, enhancing interpretability [19], [20]. In addition, the retrieval-augmented generation (RAG) approach, introduced by Lewis et al. [21], boosts knowledge extraction efficiency for large-scale data. Together, these advancements lay a robust foundation for multimodal knowledge management in froth flotation.

Despite the transformative potential of LLMs in KG construction, their application in froth flotation remains in its early stages. Existing studies in mineral processing have explored KG applications in mineral exploration and geological report analysis [6], [7], [22], yet these efforts primarily focus on textual knowledge, with limited attention to integrating image-based knowledge. In flotation processes, knowledge derived from flotation images is critical [2], [3]. However, current methods struggle to effectively integrate multimodal knowledge into the KG, lacking systematic strategies for linking images with their textual descriptions and robust quality assessment frameworks to ensure the accuracy and consistency of the resulting KG.

To meet these challenges, we propose an LLM-driven framework for constructing a multimodal KG tailored to industrial flotation processes, as shown in Fig. 1. The construction process follows the following five key steps:

1) developing a flotation knowledge tree ontology model to structure cross-procedural knowledge, encompassing plant-level operations, multicell interactions, and correlations of key parameters;
2) applying adaptive prompt optimization to improve entity and relationship extraction for domain-specific flotation data;
3) implementing a fuzzy information entropy-driven RAG vector database with dynamic chunking and optimized storage to enhance data retrieval efficiency;
4) executing a two-stage triple generation process, where initial triples are generated by a LoRA-fine-tuned lightweight LLM, followed by refinement using a large-scale LLM to produce a structured KG; and
5) storing data in Neo4j for efficient querying, dynamic web-based visualization, and quality assessment to ensure graph reliability.

This framework significantly reduces manual effort while providing robust support for knowledge management in flotation processes, facilitating efficient knowledge representation and retrieval. The main contributions of this study are as follows.

1) A flotation knowledge tree ontology is designed to serve as the structural backbone for KG construction. It organizes plant-level operations and multicell interactions, integrates multimodal flotation data, and establishes a benchmark for a domain-specific quality assessment method based on structural and semantic scoring.

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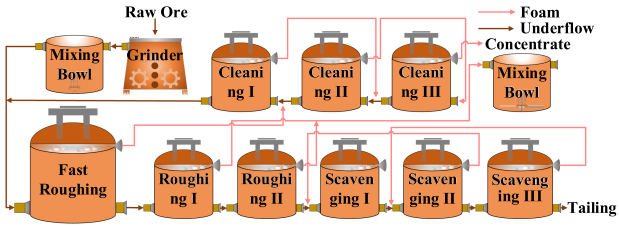


Fig. 2. Schematic diagram of the lead-zinc flotation process.

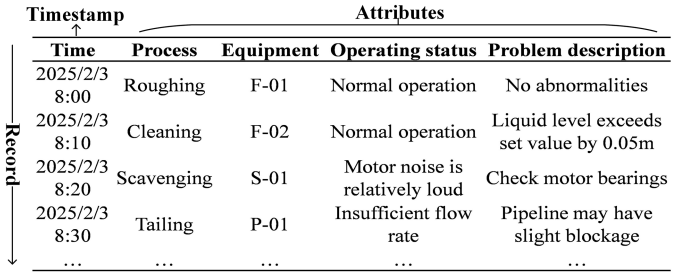


Fig. 3. Example of an inspection record from a lead-zinc froth flotation plant.

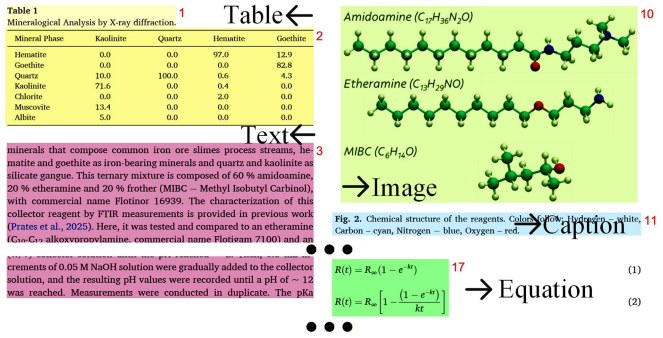


Fig. 4. Example of multimodal unstructured data from the academic literature.

2) An adaptive prompt optimization approach is proposed to iteratively refine a specialized template with flotation-specific examples and performance feedback. This mechanism empowers the LLM to extract accurate triples from complex technical narratives and heterogeneous data sources within the flotation domain.
3) A fuzzy information entropy-driven RAG method is proposed, introducing a fuzzy logic mechanism centered on domain-specific terms. It assigns weights to key terms and numerical contexts, calculates adaptive entropy for chunks, and dynamically merges segments to preserve causal relationships and semantic integrity under data uncertainty.
4) A two-stage LLM-driven KG construction pipeline is established. This pipeline generates initial triples using a LoRA fine-tuned lightweight LLM with optimized prompts and fuzzy RAG context, and a large-scale LLM refines these triples for precision and ontology alignment before mapping validated results onto the flotation knowledge tree. This approach integrates LLM reasoning, fuzzy semantic retrieval, and ontology-guided structuring into an automated, high-quality assembly for industrial flotation KG.

The rest of this article is organized as follows. Section II details the construction of the flotation dataset, including data collection and multisource data parsing. Section III presents the LLM-driven multimodal KG construction process, covering ontology modeling, LLM domain adaptation, and KG generation. Section IV discusses triple extraction experiments and KG quality assessment. Finally, Section V concludes this article.

II. Dataset Construction

Froth flotation, a critical process in mineral processing, involves multiple stages, such as fast roughing, roughing, scavenging, and cleaning [23], [24], as illustrated in Fig. 2.

The inherent complexity and data heterogeneity across these stages necessitate a comprehensive dataset for the development of a multimodal KG. The data sources collected cover key production stages, including safety inspection records, accident reports, production regulations, industry standards, equipment manuals, academic papers, and technical reports. The constructed dataset integrates structured tabular data, unstructured text, images, and semistructured data, providing a robust foundation for froth flotation-specific KGs.

A. Data Collection and Source Analysis

This section details the data collection process for domain knowledge in froth flotation, categorized into structured and unstructured data.

1) Structured Data Collection: Structured data, primarily sourced from industrial production systems, includes safety inspection records and operational logs [25]. These data are typically organized in tabular form, exemplified by inspection logs containing fields, such as timestamps, equipment names, and operational status, as shown in Fig. 3. The standardized structure can be represented by

$$D_{\text{stru}}=\{R_{si}|R_{si}=(t_i,v_{i1},v_{i2},\ldots,v_{im}),i=1,2,\ldots,n\} \tag{1}$$

where D_{stru} represents the structured dataset, R_{si} denotes the ith record, t_i is the timestamp, v_{ij} indicates the jth attribute value of the ith record, n is the total number of records, and m represents the number of attributes.

2) Unstructured Data Collection: Unstructured data, characterized by free-form text with rich content and flexible format, is primarily sourced from the literature, including industry standards, books, and academic papers. These data encompass text, formulas, charts, and images [26], [27], as illustrated in Fig. 4, forming a multimodal content collection, which can be represented by

$$\begin{aligned} D_{\text{unstru}} &= \{(C_{ui},M_{ui},A_{ui})|i=1,2,\ldots,n\} \\ M_{ui} &\in \{\text{Text},\text{Formula},\text{Table},\text{Image},\ldots\} \end{aligned} \tag{2}$$

where D_{unstru} represents the unstructured dataset, C_{ui} denotes the ith content unit, M_{ui} represents its modality type, A_{ui}

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indicates its position within the document, and n is the total number of content units.

B. Parsing and Integration of Multisource Data

To effectively leverage multisource heterogeneous data, this study employs MinerU to parse the collected PDF documents [28]. The process involves extracting text S_D, image sets I_D, and caption sets C_I, and integrating them into a unified Javascript object notation (JSON) format. The integrated result D_{JSON} is represented by

$$\begin{aligned} S_D &= s_1, s_2, \ldots, s_p,\ I_D = I_1, I_2, \ldots, I_q, \\ C_I &= c_1, c_2, \ldots, c_q \\ D_{\text{JSON}} &= \{(S_D, I_D, C_I)\} \end{aligned} \tag{3}$$

where s_i represents the ith text fragment, p is the total number of text fragments, I_j denotes the jth image, q is the total number of images, and c_j is the caption corresponding to the jth image.

This process provides a unified and efficient input foundation for the subsequent KG construction using LLMs [29].

III. LLM-Driven Multimodal KGs Construction

A. Ontology Modeling

Within the industrial context of froth flotation, ontology modeling provides a systematic framework for constructing a multilevel, cross-procedural multimodal KG. It formally models the dynamic behaviors of each flotation stage while offering a structured representation of domain concepts, entities, and their interrelationships [30], [31], [32]. To intuitively represent the hierarchical structure and process pathways, this study designs a "flotation knowledge tree," formally defined as

$$\begin{aligned} T &= (V_T, E_T, A_T) \\ V_T &= \{V_{\text{root}}, V_{\text{stage}}, V_{\text{leaf}}\} \\ E_T &= \{(v_i, v_j) | v_i, v_j \in V, i \neq j\} \\ A_T &= \{A_{\text{root}}, A_{\text{stage}}, A_{\text{leaf}}\} \end{aligned} \tag{4}$$

where V_T represents the set of nodes, encompassing root, intermediate, and leaf nodes, E_T denotes the set of edges, capturing connectivity relationships between nodes and enabling bidirectional flow from bottom-up and top-down, and A_T is the set of attributes, describing the characteristics of nodes at each level. Based on this tree structure, the ontology model further defines classes and attributes, as expressed by

$$\begin{aligned} V_{\text{root}} &= \{\text{RawOreCategory}, \ldots\}, \\ A_{\text{root}} &= \{\text{MineralComposition}, \text{ParticleSizeDistribution}, \ldots\} \\ V_{\text{stage}} &= \{\text{FastRoughingCategory}, \text{RoughingCategory}, \\ &\quad \text{ScavengingCategory}, \text{CleaningCategory}, \ldots\} \\ A_{\text{stage}} &= \{\text{ProcessStageType}, \text{EquipmentType}, \\ &\quad \text{ProcessParameters}, \ldots\} \\ V_{\text{leaf}} &= \{\text{ConcentrateCategory}, \text{TailingCategory}, \ldots\}, \end{aligned}$$

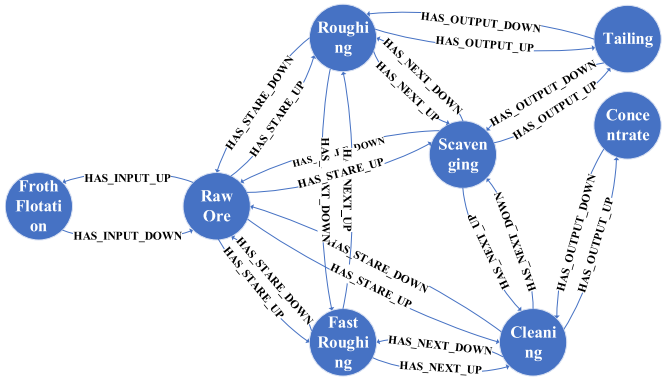


Fig. 5. Structure of the flotation knowledge tree ontology.

$$A_{\text{leaf}} = \{\text{ProductQuality}, \text{ProductionOutput}, \ldots\}. \tag{5}$$

This model comprehensively delineates the froth flotation process and its interconnections, as illustrated in Fig. 5, providing a robust foundation for the subsequent construction of the target KG.

B. LLM Domain Adaptation and Optimization

To construct a KG for froth flotation, the LLM must adapt to domain-specific characteristics [33]. This section introduces an optimization strategy that integrates adaptive prompt refinement, a specialized fuzzy RAG vector database, and LoRA-based fine-tuning of the DeepSeek model, significantly enhancing its ability to process flotation-related literature and data.

1) Adaptive Prompt Optimization: In LLM applications, the prompt serves as a directive input that guides the model toward generating targeted outputs; its quality directly determines performance in domain-specific tasks. To improve the adaptability of the LLM to the froth flotation domain, this study introduces an automated iterative prompt optimization mechanism. This mechanism refines the prompting strategy to bolster the model's comprehension of flotation knowledge while providing precise input templates for downstream tasks [34]. Leveraging dynamic feedback and the flotation knowledge tree ontology, the process follows a three-stage iteration.

a) Initial prompt generation: A structured initial prompt template, denoted as P_0, is designed based on the flotation knowledge tree. This ensures the LLM accurately recognizes domain entities and relationships. Its formal representation is given by

$$P_0 = \{R_0, C_0, E_0, O_0, T_0\} \tag{6}$$

where R_0 denotes the role definition, specifying the task objective, R_0 represents the processing principles to ensure alignment with domain standards, and R_0 and O_0 are the few-shot input and output examples, respectively, which guide the model in formatting results, and T_0 is the input text to be processed. This template generates an initial response via the LLM, constrained by the entity set V_T and relationship set E_T to ensure domain relevance.

b) Dynamic feedback assessment: The generated outputs are evaluated using metrics including precision P_{re}, recall R_{ec}, $F1$

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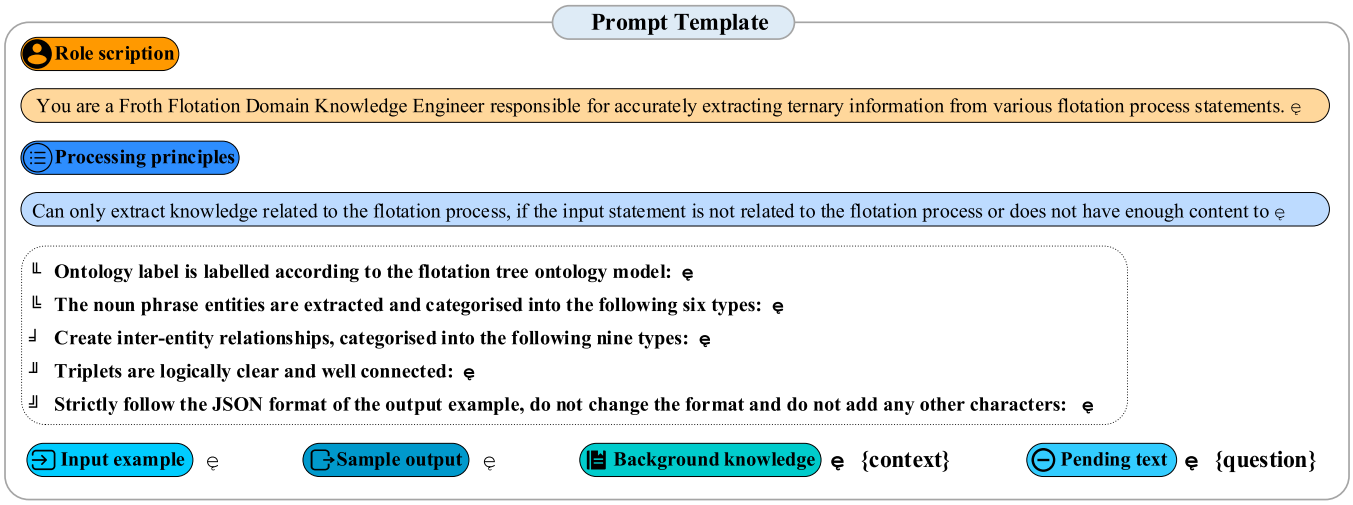


Fig. 6. Final prompt template after iterative optimization and manual integration.

score, and consistency C_{sis}. Triple matching utilizes a fuzzy matching algorithm to compute the total similarity score S_{triplet}, assigning weights of 0.3, 0.3, 0.25, and 0.15 to the similarity scores of the subject S_{subject}, object S_{object}, relationship S_{relation}, and label S_{label}, respectively. A total similarity score exceeding 0.85 is considered a valid match, as expressed by

$$P_{\text{re}} = \frac{N_{\text{valid}}}{N_{\text{pred}}}, R_{\text{ec}} = \frac{N_{\text{valid}}}{N_{\text{exp}}}$$

$$F1 = 2 \cdot \frac{P_{\text{re}} \cdot R_{\text{ec}}}{P_{\text{re}} + R_{\text{ec}}}$$

$$C_{\text{sis}} = 1 - \frac{N_{\text{conflict}}}{N_{\text{total}}}$$

$$S_{\text{triplet}}(t_1, t_2) = 0.3S_{\text{subject}} + 0.3S_{\text{object}} + 0.25S_{\text{relation}} + 0.15S_{\text{label}} \quad (7)$$

where N_{valid}, N_{pred}, and N_{exp} are the counts of valid matches, predicted triples, and expected triples, respectively; N_{conflict} is the number of logical conflicts, and N_{total} is the total triple count.

c) Iterative optimization strategy: Based on evaluation outcomes, the prompt is dynamically adjusted to produce an optimized version P_{n+1}. If $F1 < 0.6$, indicating insufficient extraction capability, an enhancement strategy decomposes complex sentence structures and supplements missing entities or relationships. If $C_{\text{sis}} < 0.8$, suggesting logical inconsistency, the optimization strengthens grammatical coherence and directional consistency among triples. If $R_{\text{ec}} < P_{\text{re}}$, indicating lower recall relative to precision, the coverage of entity and relationship sets is expanded. Each iteration updates the candidate prompt and compares it with the best-performing version; if the current $F1$ score improves, the best result is updated. The process terminates when $F1 \geq 0.90$ and $C_{\text{sis}} \geq 0.8$, or the maximum iteration limit is reached, ensuring efficient convergence [35], [36].

To validate this mechanism, the DeepSeek model was tested on representative froth flotation texts [37]. The optimized model achieved a precision, recall, and *F1* score of approximately 0.95, with a consistency score of 1.00. Notably, the peak *F1* score reached 95.06%, and the outputs closely aligned with ground truth data. This approach significantly enhances the LLM's adaptability through structured knowledge representation and dynamic feedback loops. Following refinement and consolidation, an optimized prompt template is derived, as shown in Fig. 6, fully satisfying the domain's knowledge extraction requirements.

2) Fuzzy RAG Domain Vector Database Construction: To develop a RAG system tailored for the froth flotation domain, this study proposes an enhanced vector database construction method that integrates fuzzy logic to improve vector storage efficiency and semantic retrieval accuracy [38], [39], [40]. This approach incorporates technical modules, such as text loading with automatic term weighting, fuzzy semantic chunking with entropy optimization, M3E model vectorization with fuzzy semantic weighting, vector database construction, and storage optimization. The process is systematically defined and experimentally validated, effectively capturing the domain characteristics of the froth flotation process while supporting the development of a structured knowledge system [41], [42].

a) Text loading and automatic term weighting: A fuzzy logic-based term weighting mechanism, leveraging a Gaussian function for its smooth and continuous properties, automatically identifies and prioritizes key terms to address uncertainty and noise in flotation data. Term importance is evaluated using TF-IDF scores, normalized by the maximum score ($w_{\text{score}}/w_{\text{max}}$), and mapped to a fuzzy membership value via a Gaussian function by

$$\mu(w) = \exp\left(-(w_{\text{score}}/w_{\text{max}} - 0.8)^2 / \left(2 \times 0.2^2\right)\right) \quad (8)$$

where w_{score} is the TF-IDF score of term w and w_{max} is the maximum TF-IDF score across all terms. This fuzzy approach ensures a smooth weighting of term significance. As shown in Fig. 7(a), Gaussian membership produces a smoother weight transition compared to triangular and trapezoidal functions, reducing abrupt changes in term significance. Example weights include "slurry" (1.0), "bubble" (0.882), and "concentrate grade"

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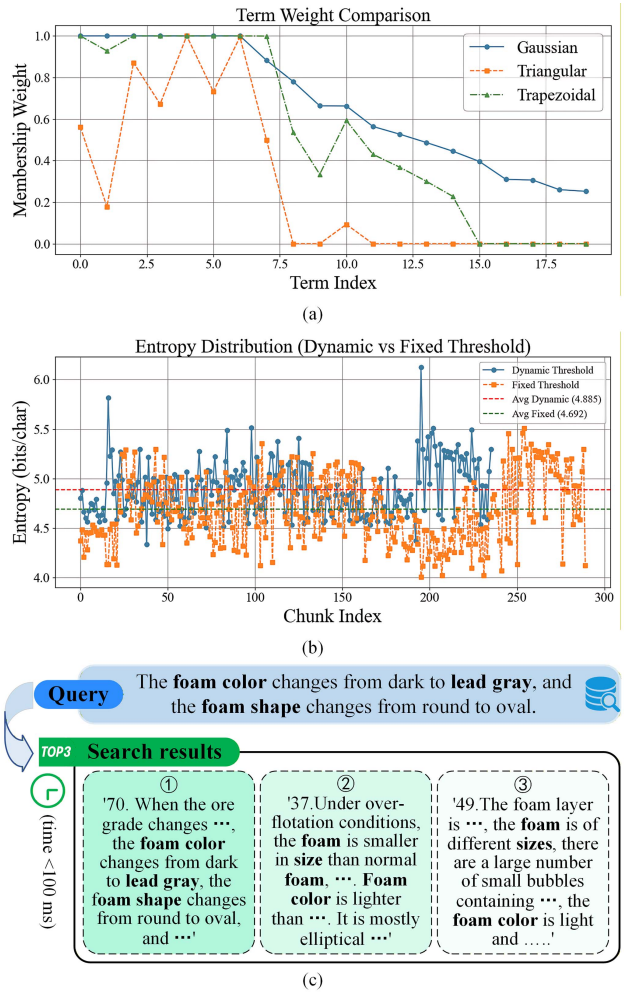


Fig. 7. Fuzzy-enhanced RAG validation. (a) Term-weight smoothness under Gaussian, triangular, and trapezoidal functions. (b) Entropy distribution with dynamic versus fixed thresholds. (c) Top-3 retrieval example.

(0.781). Key terms are selected from high TF-IDF scores via this Gaussian mapping. Dynamic stop words are filtered based on low-frequency terms to enhance domain relevance.

b) Fuzzy semantic chunking with entropy optimization: A fuzzy semantic chunking strategy balances semantic coherence and information density, using sentence boundaries as initial separators. The chunking algorithm dynamically merges blocks based on fuzzy-adjusted entropy thresholds. The fuzzy entropy of a chunk c_i is computed by

$$H(c_i) = -\sum_{w\in c_i} p(w)\log_2 p(w)\left(1+\left(0.3\left(\sum_{w\in c_i}\mu_h(w)\right)/|c_i|\right)\right) \quad (9)$$

where $p(w)$ is the normalized term frequency in the chunk and $|c_i|$ is the number of tokens in c_i. The fuzzy membership function $\mu_h(w)$ is defined as

$$\mu_h(w) = \begin{cases} \mu(w)\cdot(1.0+0.2\mathbb{I}_{\text{num}}(w)), w \in \text{terms} \\ \exp\left(-(p(w)-0.01)^2/\left(2\cdot(0.05)^2\right)\right) \\ \cdot(1.0+0.2\mathbb{I}_{\text{num}}(w)), w \notin \text{terms} \end{cases} \quad (10)$$

where $\mathbb{I}_{\text{num}}(w) = 1$ if w appears in a numerical context, and 0 otherwise; the set terms consists of high-TF-IDF domain terms identified in (8). This ensures key domain terms retain high weight, while rare terms are suppressed, with numerical context boosting relevance across both cases.

The entropy threshold is dynamically adjusted using fuzzy logic to adapt to varying information densities

$$H_{\text{min}}^{\text{fuzzy}} = H_{\text{min}}\cdot\left(1.0+0.2\exp\left(-(H(c_i)-H_{\text{min}})^2/\left(4\cdot(0.5)^2\right)\right)\right).$$

The chunking process follows a recursive merging logic:

$$c_i = \begin{cases} c_i', H(c_i') \geq H_{\text{min}}^{\text{fuzzy}} \wedge |c_i'| \leq 50 \\ c_i' + \sum_{j=1}^{k} c_{i+j}', H(c_i') < H_{\text{min}}^{\text{fuzzy}} \wedge H(c_i' + \sum_{j=1}^{k} c_{i+j}') \\ \geq H_{\text{min}}^{\text{fuzzy}} \wedge |c_i| \leq 100 \end{cases}.$$

After chunking, 236 segments of average length 48.0 characters are obtained. The first five entropy values are 4.374, 4.482, 4.209, 4.457, and 4.282 bits/char, all satisfying the adjusted threshold. As shown in Fig. 7(b), dynamic thresholds produce a more compact and consistent entropy distribution (range 4.2–5.4 bits/char, mean 4.885 bits/char) with fewer segments than a fixed threshold, ensuring higher information density and semantic stability.

c) M3E model vectorization with fuzzy semantic weighting: Segmented text c_i is encoded into 768-D embeddings using the M3E-Base model as $v_i = M3E(c_i)$, with L2 normalization $\|v_i\|_2 = 1$ ensuring unit length. To strengthen domain-specific semantic representation, cosine similarity is refined by a fuzzy membership score

$$\text{sim}_{\text{fuzzy}}(v_i, v_j) = \frac{v_i \cdot v_j}{\|v_i\|\cdot\|v_j\|}\cdot M_{\text{fuzzy}}(c_i, c_j)$$
$$M_{\text{fuzzy}}(c_i, c_j) = 1.0 + (W_{\text{term}} + B_{\text{context}})\cdot 2 \quad (11)$$

where W_{term} is the average fuzzy membership of shared domain terms between c_i and c_j, and B_{context} is the proportion of common terms relative to the total terms in c_i. This fuzzy adjustment prioritizes chunks with higher domain-specific term overlap and contextual relevance during retrieval, improving the alignment of embeddings with flotation semantics.

d) Vector database construction and storage optimization: The vector database is constructed using the Chroma framework, employing the hierarchical navigable small world (HNSW) algorithm for efficient approximate nearest-neighbor search [43]. Embeddings are indexed in batches of 100 with HNSW parameters $M = 8$ and $ef_{\text{construction}} = 40$. The construction is guided by a design objective that balances retrieval accuracy and storage efficiency, formalized as

$$L(D) = \frac{1}{N}\sum_{i=1}^{N}(1-\text{sim}_{\text{fuzzy}}(v_i, q_i))^2 + \lambda\cdot\frac{S_{\text{index}}(D)}{N} \quad (12)$$

where N is the total number of vectors, q_i represents representative query vectors, $S_{\text{index}}(D)$ denotes the index storage size, and λ is a balancing factor. The resulting index occupies 15 856 KB, with an average chunk footprint of 67.19 KB. As demonstrated

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TABLE I
LoRA Fine-Tuning Information

Configuration	Value	Description
Base Model	8B	DeepSeek-R1-Distill-Llama-8B
LoRA Rank (r)	32	Low-rank dimension
LoRA Alpha (α)	160	Scaling factor (α = 5 × r)
LoRA Dropout	0.4	Dropout rate
Target Modules	q/v_proj	Attention projections
Optimizer	AdamW	Weight decay 0.01
Initial LR	1e-4	Cosine decay to 0
Warmup Steps	100	Linear warmup
Batch Size	1	Gradient accumulation 8
Precision	BF16	Mixed precision
Epochs	15	Total steps ≈ 285
GPU Memory	48 GB	2 × RTX 4090

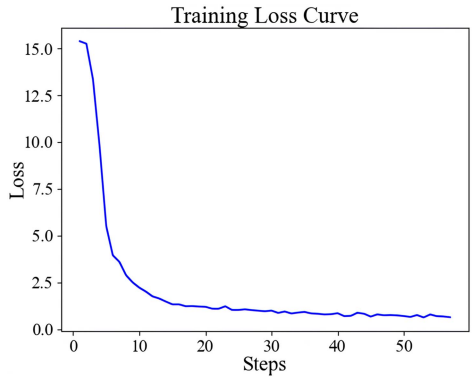


Fig. 8. LoRA fine-tuning loss curve distribution.

in Fig. 7(c), a Top-3 similarity query is executed in under 100 ms, satisfying real-time industrial requirements.

3) Domain-Specific LLM Fine-Tuning via LoRA: To enhance the LLM's knowledge representation for the froth flotation process, we fine-tune the DeepSeek-R1-Distill-Llama-8B model using LoRA [44]. This approach minimizes computational costs while preserving generalization. The process includes dataset preparation, LoRA configuration, training, and performance evaluation.

a) Domain knowledge dataset construction and preprocessing: We curate a flotation-specific dataset in JSON lines (JSONL) format. Each sample contains system prompt, user input, and expected output. To ensure input consistency, text sequences are tokenized and truncated based on a dynamic length threshold defined by

$$L_{\max} = \max(128, \min(512, 1.2 \cdot \max_{d_i}(\text{len}(\text{tokenize}(d_i))))). \quad (13)$$

The raw text sequence d_i is constructed using specific control tokens

$$d_i = [\text{SYSTEM}] \oplus s_i \oplus [\text{USER}] \oplus u_i \oplus [\text{ASSISTANT}] \oplus a_i \quad (14)$$

where s_i, u_i, a_i denote the system instruction, user query, and assistant response for the ith sample, respectively, and $\oplus$ represents string concatenation. To align with the causal language modeling objective, the tokenized sequence T_i and its corresponding label mask M_i are constructed to ensure the loss is computed solely on the target response

$$T_i = \text{tokenize}(d_i), M_i[j] = \begin{cases} \text{IGNORE}_{\text{IDX}}, & j < \text{start}_{\text{index}}(a_i) \\ T_i[j] \ , & j \geq \text{start}_{\text{index}}(a_i) \end{cases} \quad (15)$$

where $\text{IGNORE}_{\text{IDX}}$ serves as a masking token that excludes the instruction part from the loss computation, preventing the model from calculating gradients on the prompt while allowing it to attend to the full context.

b) LoRA fine-tuning configuration: LoRA is applied to query and value projection matrices in transformer layers [45]. The configuration for LoRA fine-tuning is summarized in Table I.

Training uses cosine annealing for the learning rate as

$$\eta(t) = \eta_{\min} + 0.5(\eta_{\max} - \eta_{\min})(1 + \cos(t_{\text{LoRA}}\pi / T_{\text{LoRA}})) \quad (16)$$

where $\eta_{\max}$ is the initial learning rate, $\eta_{\min}$ is the minimum learning rate, T_{LoRA} is the total steps, and t_{LoRA} is the current step.

c) Training process and performance analysis: Training runs efficiently. Logs are recorded every 5 steps, checkpoints saved every 50 steps, minimizing cross-entropy loss. Training loss decreases from 15.3803 to 0.6378, showing strong convergence, as shown in Fig. 8. The LoRA-fine-tuned model exhibits enhanced performance compared to the base model, reflecting improved domain-specific knowledge extraction capability.

C. Multimodal KG Construction

To construct a multimodal KG tailored to the froth flotation domain, we design an LLM-driven triple generation strategy that integrates an optimized prompt template with a fuzzy information entropy-based RAG mechanism. To mitigate the limitations associated with the parameter scale of the fine-tuned model, a coarse-to-fine two-stage generation strategy is employed. Initial triples are generated by the fine-tuned model using the optimized prompt and context retrieved via fuzzy RAG. Subsequently, these triples are combined with the original prompt and refined by the large-scale DeepSeek model to ensure accuracy and completeness [46], [47].

The froth flotation process generates multimodal data, including text, images (e.g., process diagrams and tables treated as images), and corresponding captions. To achieve unified knowledge representation, image data is hosted on a repository to obtain unique URLs for traceable identification. Input data is preprocessed into JSON format and handled iteratively. Utilizing the fuzzy information entropy-based RAG, relevant contextual knowledge is retrieved and paired with the optimized prompt to produce initial triples. These are then refined by the DeepSeek model to yield the final set of triples. The triple generation process is formalized as

$$T_m = f_{\text{LLM}}(x_m, f_{\text{Fine}_{\text{LLM}}}(x_m, C_{\text{RAG}})), m \in \{\text{text}, \text{caption}\} \quad (17)$$

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where x_m represents the input data, C_{RAG} denotes the context retrieved via fuzzy RAG, and T_m is the set of generated triples.

These triples form a connected subgraph for each input segment. To facilitate semantic organization, the entity with the highest degree within the subgraph is identified as the representative node v^*. This node serves as the anchor for linking the extracted instance to the ontology backbone and, where applicable, to the source image

$$v^* = \arg\max_{v \in S} \deg(v) \tag{18}$$

where S is the set of entity nodes in the subgraph formed by T_m, and $\deg(v)$ represents the node degree.

The KG is constructed using the flotation knowledge tree ontology as its backbone. Mathematically, the KG is represented as a directed graph $G = (V, E)$ [48]. The edge set E integrates the extracted triples, the hierarchical ontology links, and the multimodal associations:

$$E = E_{\text{triples}} \cup E_{\text{img}} \cup E_{\text{onto}} \tag{19}$$

where E_{triples} is the union of all generated triples T_m, E_{onto} represents the logical connection between the extracted data and the ontology backbone, linking every representative node v^* to its corresponding ontology concept b

$$E_{\text{onto}} = \{(b, \text{HAS_ENTITIES}, v^*) | b = M(v^*)\} \tag{20}$$

where $M(v^*)$ maps v^* to the specific ontology node b. Finally, E_{img} establishes the multimodal link specifically for triples derived from image captions

$$E_{\text{img}} = \{(v^*, \text{HAS_IMG}, u) | v^* \in V_{\text{caption}}\} \tag{21}$$

where u is the image URL and V_{caption} denotes the set of representative nodes derived from image caption data.

This method extracts triples using the LLM with the optimized prompt and fuzzy RAG context, then assembles them into a multimodal KG with the flotation knowledge tree as its backbone. Image associations improve interpretability, supporting deep integration and structured representation of multimodal data.

IV. Experiments and KG Quality Assessment

A. Triple Extraction Experiments

To construct a multimodal KG tailored for the froth flotation domain, this study develops methods integrating the flotation knowledge tree ontology (see Section III-A) with LLM domain adaptation techniques (see Section III-B). The method includes the following:

1) optimizing prompts iteratively to produce an optimized prompt template, achieving precision of 0.95, recall of 0.95, $F1$ score of 0.95, and consistency of 1.00 (see Section III-B-1);
2) building a fuzzy logic-based RAG vector database for semantic retrieval, achieving query response time below 100 ms with consistent entropy distribution (see Section III-B-2);
3) fine-tuning the LLM with LoRA, reducing loss from 15.3803 to 0.6378 (see Section III-B-3);
4) applying a two-stage triple generation process, where the fine-tuned LLM generates initial triples, refined by the large-scale LLM (see Section III-C).

These steps collectively enable the construction of a multi-level, cross-procedural KG for the froth flotation domain.

To evaluate the effectiveness of the proposed triple extraction approach before full KG assembly, we curate a domain-specific dataset of 100 flotation process sentences. Each sentence is manually annotated with ground-truth triples by domain experts to ensure accuracy. Statistics of the dataset are shown in Table II.

TABLE II
Statistics of the Dataset

Metric	Value	Metric	Value
Total Samples	100	Total Triples	375
Avg. Triples	3.75	Avg. Text Length	75 chars
Max Triples	10	Max Length	118 chars
Min Triples	1	Min Length	16 chars

TABLE III
Comparative Results on the Dataset

Method	Recall	Precision	F1-score
Proposed	**0.9467**	**0.7717**	**0.8503**
LLM-FewShot	0.5333	0.4274	0.4745
UIE	0.2960	0.0960	0.1450
Rule + LTP	0.1893	0.1068	0.1365

Note: The bold entities indicate that the method proposed by the authors of this article achieves the best performance.

We evaluated the effectiveness of our proposed method against three representative extraction paradigms:

Rule + language technology platform (LTP) [49]: A traditional rule-based approach utilizing the LTP for dependency parsing. Triples are extracted based on rigid syntactic patterns, e.g., subject–verb–object paths.

Universal information extraction (UIE) [50]: A state-of-the-art supervised method from the pre-LLM era, employing a Schema-guided Seq2Seq generation framework based on pre-trained language models.

LLM-FewShot [51]: A generative approach representing the general LLM paradigm, where extraction is framed as a multi-turn conversational query without domain-specific fine-tuning.

The comparative performance was evaluated using precision, recall, and $F1$-score [52]. The results, averaged over the 100 samples, are presented in Table III.

As demonstrated in Table III, the proposed method significantly outperforms all baselines, achieving the highest *F1*-score of 0.8503. While LLM-FewShot demonstrates moderate capability, it suffers from lower precision due to hallucinated relations and inconsistent formatting. Traditional methods, specifically UIE and Rule + LTP, perform poorly in this specialized domain, with *F1*-scores below 0.15. This highlights their inability to handle the complex semantic logic and domain-specific context inherent in flotation processes.

To further analyze the stability of extraction, Fig. 9 visualizes the trend of correctly extracted triples versus the ground-truth count across document IDs. The visualization confirms that the proposed method maintains high consistency across varying text

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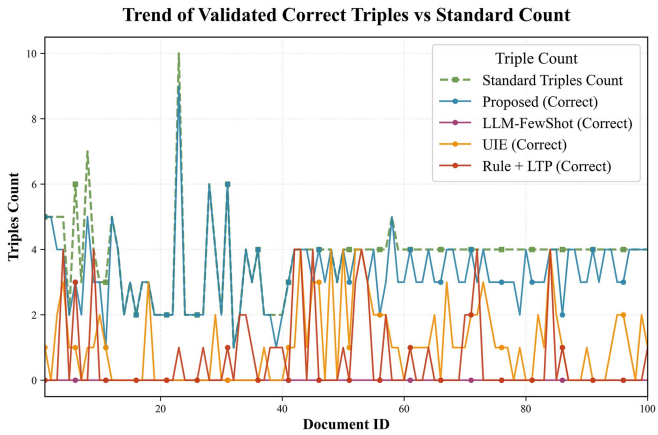


Fig. 9. Comparison of validated vs. ground-truth triples on the dataset.

TABLE IV
COMPARISON OF TRIPLE EXTRACTION ACROSS METHODS

Input: If flotation time is kept constant, then as pulp concentration increases, the production efficiency of the flotation machine also increases accordingly.

Method	Extracted Triples	Triple Count	Domain Alignment
Proposed	(flotation time, keep→constant, pulp concentration), (pulp concentration, increase→improve, production efficiency of flotation machine)	2	High
LLM-FewShot	(pulp concentration, increase→cause, production efficiency improvement), (flotation time, keep→constant, production efficiency improvement)	2	Medium
UIE	(flotation machine, improve, production efficiency)	1	Low
Rule + LTP	(if keep, change/state, change), (flotation time, change/state, change), (production efficiency of flotation machine, change/state, improve)	3	Low

lengths and complexities, whereas baseline methods frequently miss key relations or introduce noise.

To visually illustrate the differences in extraction, a case study of a typical sentence is presented in Table IV.

As shown in the case study, the proposed method consistently produces accurate, domain-aligned triples with standardized terminology and correct causality. LLM-FewShot often generates verbose or loosely connected relations. UIE yields fragmented phrases that lack the full triple structure. Rule + LTP is highly sensitive to syntactic variations, resulting in spurious or incorrect relations.

These results confirm that our approach significantly outperforms existing methods in extracting reliable, domain-coherent triples from flotation text. We use the data prepared in Section II to construct a multimodal KG for the flotation domain following the process in Section III-C. The quality of the resulting graph is evaluated in the following subsection.

B. KG Quality Assessment

Evaluating the quality of KGs is critical for ensuring their reliability and applicability in industrial contexts, particularly for flotation processes [53], [54]. This study utilizes a flotation knowledge tree ontology to assess quality across two dimensions: topological structure and semantic representation. By analogizing the flotation process to a fruit tree, the topological structure captures the graph's connectivity, resembling the tree's external framework, while semantic representation reflects the intrinsic accuracy, akin to the fruit's inner quality. This analogy incorporates flotation industry characteristics, enabling systematic, and comprehensive quality quantification.

1) Structural Assessment: The flotation knowledge tree ontology serves as the backbone of the KG, providing a robust framework that supports multilevel, cross-procedural optimization [55]. To quantify structural integrity and verify alignment with this ontological framework, we developed six metrics by integrating topological theory with the industrial features of froth flotation. These metrics evaluate connectivity, process integrity, dynamic balance, knowledge density, process depth, and feedback mechanisms.

Isolated node ratio I_s: Rooted in topological connectivity theory, this metric measures the proportion of disconnected nodes. Minimizing I_s ensures the effective integration of knowledge points and prevents information silos. It is calculated as

$$I_{\mathrm{s}} = \frac{N_{\mathrm{isolated}}}{N_{\mathrm{total}}} \tag{22}$$

where N_{isolated} is the count of nodes with a degree of zero and N_{total} is the total number of nodes. An I_{s} value of 0 indicates full connectivity.

Process coverage rate C_{s}: Tailored to the flotation hierarchy, this metric verifies the inclusion of core process stages (i.e., fast roughing, roughing, scavenging, and cleaning), ensuring the KG covers the complete workflow from raw ore to tailings [54]. It is defined as

$$C_{\mathrm{s}} = \frac{N_{\mathrm{covered}}}{N_{\mathrm{core}}} \tag{23}$$

where N_{covered} is the number of identified core process nodes and N_{core} is the total number of required stages. A value of 1 signifies complete process coverage.

Bidirectional relationship balance B_s: This metric leverages topological path symmetry to assess the equilibrium between forward process flows and feedback loops within the flotation knowledge tree [56]

$$B_s = 1 - \left| \frac{N_{\mathrm{down}} - N_{\mathrm{up}}}{N_{\mathrm{down}} + N_{\mathrm{up}} + 1} \right| \tag{24}$$

where N_{down} and N_{up} represent the counts of downstream and upstream relationships, respectively. A value near 1 indicates a balanced bidirectional mechanism.

Process node knowledge coverage Z_s: Drawing on node degree distribution theory, Z_s quantifies the richness of knowledge associated with specific process nodes, mitigating uneven

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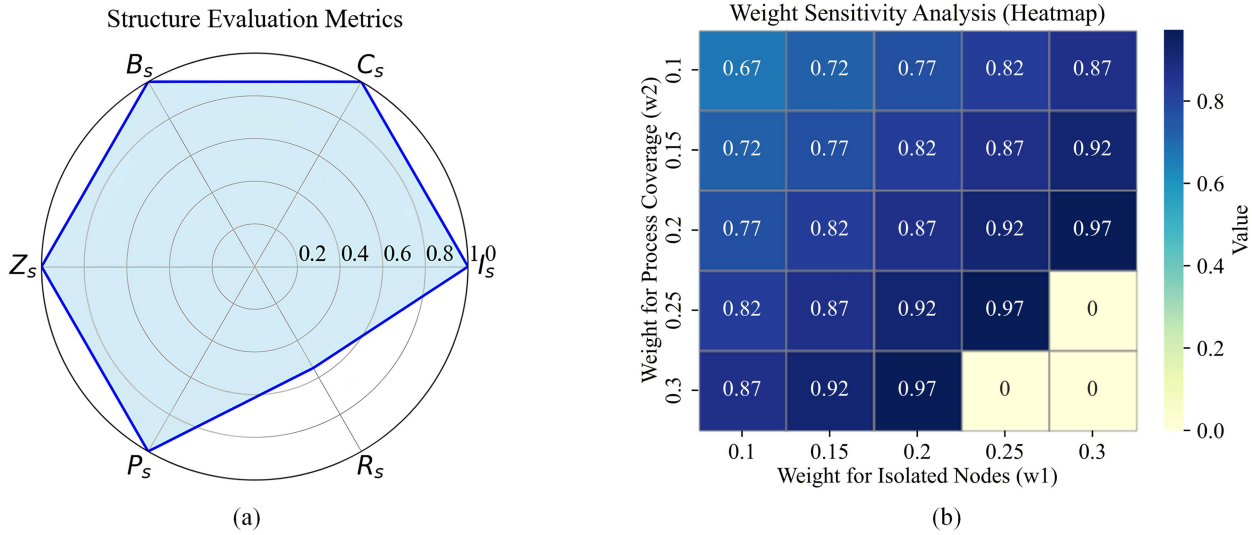


Fig. 10. Comprehensive analysis of structure evaluation metrics for the froth flotation KG. (a) Radar chart showing structure evaluation metrics scores. (b) Weight sensitivity heatmap analyzing the impact of weight variations on the score.

knowledge distribution [57]

$$Z_s = \sigma\left(\frac{\bar{N}_{\text{stage}} - \mu}{\delta}\right) \tag{25}$$

where $\bar{N}_{\text{stage}}$ is the average degree of process nodes, μ and δ are the global mean and standard deviation of node degrees, and $\sigma(\cdot)$ is the Sigmoid function. A value approaching 1 denotes sufficient knowledge density.

Longest process path length P_s: This metric evaluates the logical depth of the process from raw ore to concentrate, ensuring comprehensive stage coverage while avoiding redundancy [58]

$$P_s = \min\left(1, \frac{L_{\max}}{L_{\text{optimal}}}\right) \tag{26}$$

where $L_{\max}$ is the longest path length in the graph and L_{optimal} is the standard industrial path length.

Process loop rationality R_{s}: Based on directed cycle analysis, this metric measures the proportion of process-related loops, optimizing flotation recovery feedback mechanisms and minimizing interference from irrelevant loops [53], calculated by

$$\begin{aligned} D_s &= \frac{N_{\text{process}}}{N_{\text{edges}}}, R_{\text{s}} \\ &= \sigma\left(\frac{T_{\text{mid}}}{T_{\min} + |D - T_{\text{target}}|}\right) \cdot e^{-\max(0, |D - T_{\max}|)} \end{aligned} \tag{27}$$

where D_s represents normalized loop density, N_{process} is the number of process-related loops, and N_{edges} is the total edge count. Based on statistical feedback loop distributions in flotation processes and industrial design experience [59], we define the feedback density range using thresholds derived from edge counts (e.g., $T_{\min} \approx 20/N_{\text{edges}}$,$T_{\max} \approx 60/N_{\text{edges}}$). The formula incorporates a Sigmoid component for the target range and an exponential decay for excessive deviations.

The structural evaluation was conducted on the constructed KG, which comprises 3728 nodes and 5920 edges. As illustrated in Fig. 10(a), the resulting metric scores are: $I_s = 0.0000$, $C_{\text{s}} = 1.0000$, $B_s = 1.0000$, $Z_s = 0.9996$, $P_s = 1.0000$, and $R_{\text{s}} = 0.5488$.

These coupled metrics enable multilevel evaluation: I_s and C_{s} establish the foundation, B_s deepens the structure, Z_s enriches knowledge, and P_s and R_{s} enhance feedback, fitting the flotation process. The comprehensive structural quality score S_1 is calculated as

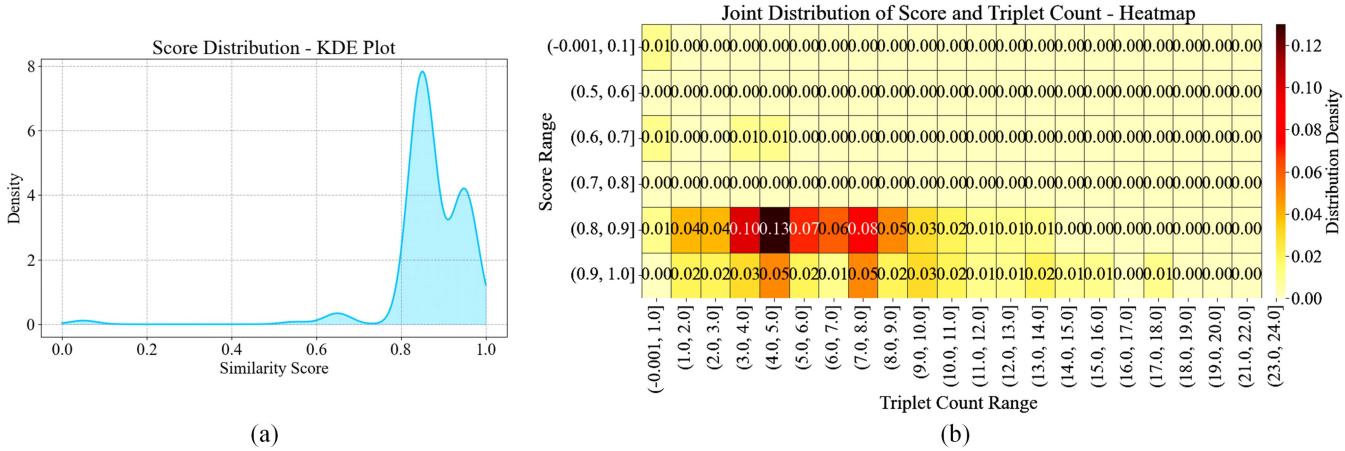
$$\begin{aligned} S_1 = {} & 0.25(1 - I_{\text{s}}) + 0.25C_s + 0.2B_s + 0.15Z_s \\ & + 0.1P_s + 0.05R_s \end{aligned} \tag{28}$$

where I_s and C_{s} are assigned the highest weights, as connectivity and process completeness are fundamental prerequisites for a functional KG, where knowledge silos must be strictly avoided. To validate this weighting strategy, the weight sensitivity heatmap in Fig. 10(b) analyzes the impact of varying the weights of the isolated node ratio and process coverage rate on the final score. The observed positive correlation between weight increases and the overall score confirms the rationality of the assigned weights.

Ultimately, the KG achieves a comprehensive structure score of $S_1 = 0.9774$, indicating an exceptional topological structure that effectively supports downstream applications.

2) Semantic Assessment: The semantic quality of the KG in the froth flotation domain significantly influences the accurate representation of process knowledge. Inconsistencies can lead to misinterpretations of entity relationships, compromising downstream applications. To validate semantic fidelity, this study introduces a reverse verification method utilizing an LLM to evaluate the consistency between triples extracted from the original text and their source sentences.

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(a) (b)

Fig. 11. Comprehensive analysis of semantic assessment results. (a) Kernel density estimation plot showing similarity score distribution. (b) Heatmap indicating the joint distribution of scores and triple counts.

The extraction process records data as $\{\text{text}, T_m, n_{\text{count}}\}$, where text is the original sentence, $T_m = \{(s_i, r_i, o_i)\}$ is the extracted triple set, and n_{count} denotes the number of triples. To ensure evaluation efficiency and representativeness, we developed a dynamic sampling strategy incorporating the Fibonacci sequence and the golden section ratio [58].

Dynamic sampling strategy: The sampling process is designed to balance efficiency and coverage across varying sentence complexities. The total record count N is processed in batches of size $B = 100$, targeting a base sample size s with a proportion $\rho = 0.2$. The strategy also dynamically adjusts s based on batch record counts. To handle complexity, sentences are classified as complex or simple based on a triple count threshold c_ϕ, which is determined by the ϕ-quantile of the triple count distribution within the batch, where $\phi \approx 0.618$ represents the golden section ratio. The sample quota for complex sentences is calculated as $s_c = \min(s \cdot \phi, \sum I(c_j \geq c_\phi))$, with the remainder allocated to simple sentences $s_s = s - s_c$, where c_j represents the number of triples in the j_{th} record. Furthermore, to ensure diverse structural representation, triple counts are segmented into intervals, and sampling weights are assigned using the Fibonacci sequence $F = [1, 2, 3, 5, 8, 13]$, where the probability for each interval corresponds to $p_i = F_i / \sum F$.

LLM-based Scoring: semantic scoring is conducted by an LLM, with inputs comprising the original sentence $text$ and the triple set T_m, producing a consistency score $\in [0, 1]$. The scoring function is expressed as score $= f_{\text{LLM}}(\text{prompt}(\text{text}, T_m; \theta))$, where θ represents the prompt parameters. The output includes both a numerical score and a reasoning rationale. Samples with $s < 0.5$ are classified as low-quality. The overall semantic score S_2 incorporates a variance adjustment to assess distribution stability, mitigating over reliance on the mean

$$S_2 = \bar{s} + \sigma_s \cdot \sqrt{\frac{1}{N_s - 1} \sum_{i=1}^{N_s} (s_i - \bar{s})^2}, \bar{s} = \frac{1}{N_s} \sum_{i=1}^{N_s} s_i \quad (29)$$

where N_s is the total sample count, $\bar{s}$ is the mean score, and σ_s is the variance weight.

The experiment involved 1656 records, from which 331 samples were selected. As illustrated in Fig. 11(a), the probability density of similarity scores is predominantly concentrated between 0.8 and 1.0, indicating strong semantic consistency. Fig. 11(b) visualizes the joint distribution of scores and triple counts, where high-density regions cluster in the 0.8–1.0 score range and 1–5 triple count interval, confirming the model's robust extraction capability. With 96.68% of samples scoring above 0.8, the overall semantic score is calculated as $S_2 = 0.8809$.

The final comprehensive evaluation integrates scores from both structural and semantic dimensions using a weighted average approach

$$S_{\text{Final}} = 0.4S_1 + 0.6S_2. \quad (30)$$

Substituting the structural score $S_1 = 0.9774$ and semantic score $S_2 = 0.8809$, the final quality assessment score is $S_{\text{Final}} = 0.9195$. This result demonstrates that the KG excels in both structural integrity and semantic consistency, offering robust support for knowledge representation in the froth flotation domain.

C. Knowledge Storage and Web Visualization

The constructed KG is stored in Neo4j, a high-performance graph database optimized for relational data management. While Neo4j efficiently handles graph topology, it lacks native support for visualizing multimodal assets, such as images. To address this, we developed a web-based visualization platform utilizing the Django framework integrated with D3.js. This platform dynamically retrieves graph data in JSON format and renders it using a force-directed layout algorithm to ensure balanced node distribution. The interface features color-coded entity categorization and interactive navigation functions, including zooming and node dragging. Crucially, it supports multimodal data integration by rendering image nodes as thumbnails, allowing direct verification of visual assets, such as froth images and equipment diagrams. As shown in Fig. 12, this platform provides a robust

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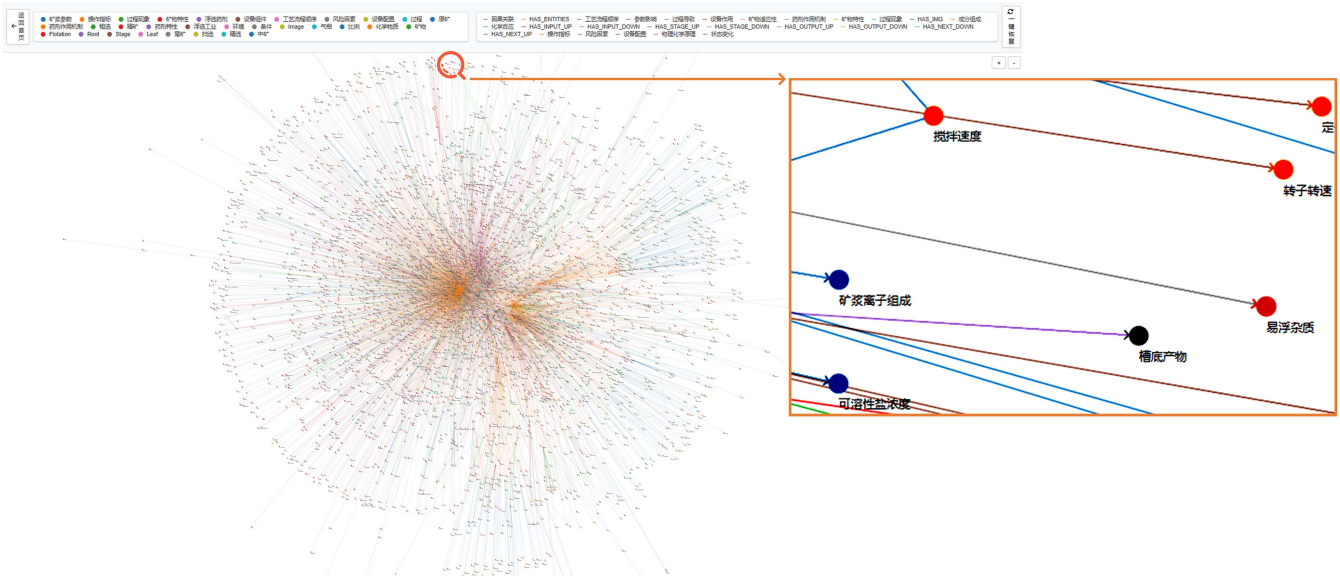


Fig. 12. KG: nodes color-coded by type, arrows show direction, image nodes as thumbnails, top legend lists types, with zoom and drag.

interface for the intuitive management and analysis of industrial knowledge.

D. KG in Practice

While the constructed multimodal KG serves as a systematic knowledge repository for plant operators, its current utility is primarily retrieval-based. To extend its capability towards dynamic problem-solving, future work will focus on integrating the KG with LLMs to develop an intelligent question-answering system.

By leveraging the structural knowledge of the KG and the reasoning capability of LLMs, this system will assist operators in addressing complex operational queries, such as identifying the "optimal reagent dosage for pH 5.2" and diagnosing "causes of low recovery." The system aims to understand domain-specific queries and provide real time, actionable insights for process adjustments. As the system evolves, further integration with plant control systems is envisioned to enable proactive capabilities, including automated anomaly detection, predictive maintenance, and optimized decision-making. Ultimately, this KG-augmented intelligent system will drive operational efficiency, reduce downtime, and facilitate the automation of flotation processes.

V. Conclusion

This study presents an LLM-driven framework for constructing KGs tailored to industrial flotation processes. By developing the flotation knowledge tree ontology, we established a hierarchical backbone that organizes plant-level operations and multicell interactions, facilitating structured representation and domain-specific quality assessment. To overcome data scarcity and complexity, the framework integrates iterative prompt optimization, fuzzy information entropy-based RAG, and LoRA fine-tuning, which collectively enhance the adaptability of LLMs to flotation-specific data. Furthermore, a coarse-to-fine two-stage generation strategy—combining a fine-tuned model for initial extraction with a large-scale model for refinement—automates the construction of high-quality multimodal KGs. In industrial settings, the framework processes flotation data to build graphs for operational optimization and decision support, advancing efficient knowledge management in flotation.

Despite its effectiveness, the framework's generalization to other domains requires redesigning the ontology and LLM adaptation, necessitating additional development effort. High computational costs of the large-scale model may limit accessibility, although API-based services could mitigate this. Noisy data may reduce graph quality, and static graphs challenge dynamic processes. Future work will explore API integration, incremental learning, and applying graphs to practical conditions for intelligent process management.

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